

We said that Alice and Bob want to communicate and wish to do so “securely,” but what precisely does this mean? As we will see, security (like love) is a many-splendored thing; that is, there are many facets to security. Certainly, Alice and Bob would like for the contents of their communication to remain secret from an eavesdropper. They probably would also like to make sure that when they are communicating, they are indeed communicating with each other, and that if their communication is tampered with by an eavesdropper, that this tampering is detected. In the first part of this chapter, we’ll cover the fundamental cryptography techniques that allow for encrypting communication, authenticating the party with whom one is communicating, and ensuring message integrity.

In the second part of this chapter, we’ll examine how the fundamental cryptography principles can be used to create secure networking protocols. Once again taking a top-down approach, we’ll examine secure protocols in each of the (top four) layers, beginning with the application layer. We’ll examine how to secure e-mail, how to secure a TCP connection, how to provide blanket security at the network layer, and how to secure a wireless LAN. In the third part of this chapter we’ll consider operational security, which is about protecting organizational networks from attacks. In particular, we’ll take a careful look at how firewalls and intrusion detection systems can enhance the security of an organizational network.

8.1 What Is Network Security?

Let’s begin our study of network security by returning to our lovers, Alice and Bob, who want to communicate “securely.” What precisely does this mean? Certainly, Alice wants only Bob to be able to understand a message that she has sent, even though they *are* communicating over an insecure medium where an intruder (Trudy, the intruder) may intercept whatever is transmitted from Alice to Bob. Bob also wants to be sure that the message he receives from Alice was indeed sent by Alice, and Alice wants to make sure that the person with whom she is communicating is indeed Bob. Alice and Bob also want to make sure that the contents of their messages have not been altered in transit. They also want to be assured that they can communicate in the first place (i.e., that no one denies them access to the resources needed to communicate). Given these considerations, we can identify the following desirable properties of **secure communication**.

- *Confidentiality*. Only the sender and intended receiver should be able to understand the contents of the transmitted message. Because eavesdroppers may intercept the message, this necessarily requires that the message be somehow **encrypted** so that an intercepted message cannot be understood by an interceptor. This aspect of confidentiality is probably the most commonly perceived

meaning of the term *secure communication*. We'll study cryptographic techniques for encrypting and decrypting data in Section 8.2.

- *Message integrity*. Alice and Bob want to ensure that the content of their communication is not altered, either maliciously or by accident, in transit. Extensions to the checksumming techniques that we encountered in reliable transport and data link protocols can be used to provide such message integrity. We will study message integrity in Section 8.3.
- *End-point authentication*. Both the sender and receiver should be able to confirm the identity of the other party involved in the communication—to confirm that the other party is indeed who or what they claim to be. Face-to-face human communication solves this problem easily by visual recognition. When communicating entities exchange messages over a medium where they cannot see the other party, authentication is not so simple. When a user wants to access an inbox, how does the mail server verify that the user is the person he or she claims to be? We study end-point authentication in Section 8.4.
- *Operational security*. Almost all organizations (companies, universities, and so on) today have networks that are attached to the public Internet. These networks therefore can potentially be compromised. Attackers can attempt to deposit worms into the hosts in the network, obtain corporate secrets, map the internal network configurations, and launch DoS attacks. We'll see in Section 8.9 that operational devices such as firewalls and intrusion detection systems are used to counter attacks against an organization's network. A firewall sits between the organization's network and the public network, controlling packet access to and from the network. An intrusion detection system performs “deep packet inspection,” alerting the network administrators about suspicious activity.

Having established what we mean by network security, let's next consider exactly what information an intruder may have access to, and what actions can be taken by the intruder. Figure 8.1 illustrates the scenario. Alice, the sender, wants to send data to Bob, the receiver. In order to exchange data securely, while meeting the requirements of confidentiality, end-point authentication, and message integrity, Alice and Bob will exchange control messages and data messages (in much the same way that TCP senders and receivers exchange control segments and data segments). All or some of these messages will typically be encrypted. As discussed in Section 1.6, an intruder can potentially perform

- *eavesdropping*—sniffing and recording control and data messages on the channel.
- *modification, insertion, or deletion* of messages or message content.

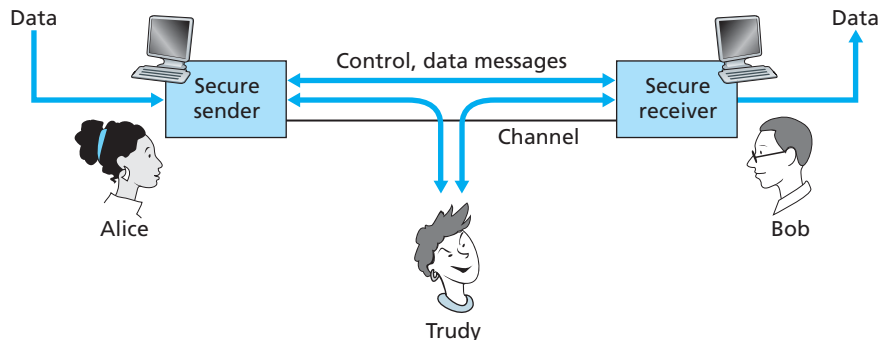


Figure 8.1 ♦ Sender, receiver, and intruder (Alice, Bob, and Trudy)

As we'll see, unless appropriate countermeasures are taken, these capabilities allow an intruder to mount a wide variety of security attacks: snooping on communication (possibly stealing passwords and data), impersonating another entity, hijacking an ongoing session, denying service to legitimate network users by overloading system resources, and so on. A summary of reported attacks is maintained at the CERT Coordination Center [CERT 2012].

Having established that there are indeed real threats loose in the Internet, what are the Internet equivalents of Alice and Bob, our friends who need to communicate securely? Certainly, Bob and Alice might be human users at two end systems, for example, a real Alice and a real Bob who really do want to exchange secure e-mail. They might also be participants in an electronic commerce transaction. For example, a real Bob might want to transfer his credit card number securely to a Web server to purchase an item online. Similarly, a real Alice might want to interact with her bank online. The parties needing secure communication might themselves also be part of the network infrastructure. Recall that the domain name system (DNS, see Section 2.5) or routing daemons that exchange routing information (see Section 4.6) require secure communication between two parties. The same is true for network management applications, a topic we examine in Chapter 9. An intruder that could actively interfere with DNS lookups (as discussed in Section 2.5), routing computations [RFC 4272], or network management functions [RFC 3414] could wreak havoc in the Internet.

Having now established the framework, a few of the most important definitions, and the need for network security, let us next delve into cryptography. While the use of cryptography in providing confidentiality is self-evident, we'll see shortly that it is also central to providing end-point authentication and message integrity—making cryptography a cornerstone of network security.

8.2 Principles of Cryptography

Although cryptography has a long history dating back at least as far as Julius Caesar, modern cryptographic techniques, including many of those used in the Internet, are based on advances made in the past 30 years. Kahn's book, *The Codebreakers* [Kahn 1967], and Singh's book, *The Code Book: The Science of Secrecy from Ancient Egypt to Quantum Cryptography* [Singh 1999], provide a fascinating look at the long history of cryptography. A complete discussion of cryptography itself requires a complete book [Kaufman 1995; Schneier 1995] and so we only touch on the essential aspects of cryptography, particularly as they are practiced on the Internet. We also note that while our focus in this section will be on the use of cryptography for confidentiality, we'll see shortly that cryptographic techniques are inextricably woven into authentication, message integrity, nonrepudiation, and more.

Cryptographic techniques allow a sender to disguise data so that an intruder can gain no information from the intercepted data. The receiver, of course, must be able to recover the original data from the disguised data. Figure 8.2 illustrates some of the important terminology.

Suppose now that Alice wants to send a message to Bob. Alice’s message in its original form (for example, “Bob, I love you. Alice”) is known as **plaintext**, or **cleartext**. Alice encrypts her plaintext message using an **encryption algorithm** so that the encrypted message, known as **ciphertext**, looks unintelligible to any intruder. Interestingly, in many modern cryptographic systems, including those used in the Internet, the encryption technique itself is *known*—published, standardized, and available to everyone (for example, [RFC 1321; RFC 3447; RFC

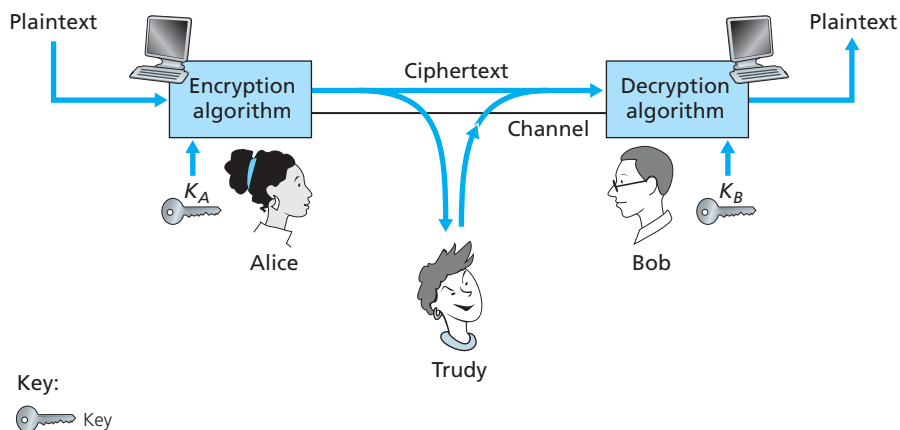


Figure 8.2 ♦ Cryptographic components